Fantasy Football Projection Model using Machine Learning

Daniel Beckley

Abstract

This self-conducted project aims to construct a fantasy football projection system, predicting per-game counting statistics for Quarterbacks (QBs), Running Backs (RBs), Wide Receivers (WRs), and Tight Ends (TEs). These statistics will then be converted to fantasy points and presented in rankings by position. Researching relevant sources will be the basis in constructing datasets used to train various machine learning models on, including Lasso, Ridge, Gradient Boosting, and Mixed Effects models. The full project and results can be found here.

1. Introduction and Motivation

Fantasy Football is a popular game amongst sports fans involving the management of individually-picked rosters of NFL players, whose game statistics are converted to fantasy points that dictate the outcomes of head-to-head matchups between users. My friends and I have been avid fantasy football players since 2018, sparking competitiveness, banter, and lasting memories. In middle, high school, and beyond, fantasy football and sports in general have played a pivotal role in maintaining our friendships that have since stretched geographically. They have driven passionate conversations and united us in our unwavering support of the Bay Area (Warriors, 49ers, Giants) and our praying for the downfall of SoCal sports teams. Combining this passion with an initiative tailored to build future skills will allow me to set myself up for my career in a more personally cultivated manner than methods otherwise. While I have varying potential career paths like data science, finance, and research, a project like this one will bode well for my future regardless of the one I choose. This is meant to serve as a learning process of trial and error, one exposing me to applicable industry practices outside the scope of my academics. This further serves as a prolonged, applicable exposure to machine learning and the Python programming language, which I previously had limited experience with (three courses between the two). My remaining prior programming experience mostly entailed data structures and algorithms in C/C++, so taking a direction in this path and continuing to do so in future courses (upper division Machine Learning, Deep Learning, Database Systems) will allow me to cater my skillset towards the movement of the fields I wish to pursue.

I would also like to be transparent about my usage of AI, sourcing my chat history and mentioning the areas I utilized it with as they come up in this report. Unless specified, idea generation was entirely provided by myself, including the research of relevant statistics, feature mapping, visualizations, and others, alongside interpretations and findings in the data. Where it especially came to benefit was as a learning tool, particularly for programming conventions, explaining more abstract data cleaning concepts, and implementing unfamiliar machine learning

models. I was able to ask questions about varying steps of the process to gain a deeper understanding of this project's functions. I was also able to correct some misclarifications that didn't align with my particular approach for this project, ensuring my vision held true and that I remained in the driver's seat.

2. Data

2.1. Constructing The Positional Datasets

I researched and pulled information from public, free online sources to create an extensive dataset for each of the positions of QB, RB, WR, and TE. FantasyPros (n.d.) provided the bulk of the datasets, with its vast collection of data and simple conversion to clean Excel spreadsheets. NFL Next Gen Stats (n.d.), ESPN Analytics (n.d.), and the nflreadpy library (Carl & Ho, 2025) served as supplementary, yet useful sources of advanced statistics. Since NFL Next Gen Stats had meaningful underlying statistics yet only data from 2018 onwards, that range became the one I used for this project. I aimed to use appropriate metrics to accurately predict 17 game season standard scoring statistics to be converted to fantasy points. Some highlighted metrics and brief reasoning for their inclusion can be broken down by category (general, passing, rushing, receiving) and source below:

General:

- Fantasy Pros (n.d.):
    - Snap analysis (Snaps/G, Touch%, Rush %, Rec%): relative measure of playing time and usage rates, which were omitted for QBs due to their domination of respective positional Snap %. Both are potentially indicative of more stable volume roles.
    - Touchdown Rate (Actual vs. Adjusted): percentage of attempts (respective to attempt type - passing, rushing, receiving) resulting (actual) or expected to result (adjusted) in touchdowns. Differences between the two can be particularly helpful in prediction touchdown regression or can be a measure of touchdown ability relative to the average player if the trends persist.
    - Redzone Statistics (Volume, Usage Rates, Scoring): how productive or busy a player is in the Redzone can drive touchdown counts.
- nflreadpy (Carl & Ho, 2025):
    - Draft (Round, Pick): potential volume and general performance indicator based on statistical projectability out of college and front office draft investment into the player. Undrafted players were filled with a round of 8 and a pick of 300 for a deducted but reasonable quantification.

    - Contracts (Inflation Adjusted APY, Contract Duration, % Guaranteed at Signing): potential volume and general performance indicator based on financial commitment rewarding past production.
    - 2026 Week 1 Starters: loaded depth charts from week 1 (09/09/26) of the 2026 NFL season to create a bank of relevant players to project. Filtered depth charts to QB1s, RB1/2s, WR1-4s, TE1/2s, and K1s to account for players expected to have meaningful roles for the target season.
- Spotrac (n.d.):
    - Significant injuries: on personal judgment (scanned for significant missed games, acl, achilles, high ankle injuries) manually entered a boolean indicator for significant injuries by accessing Spotrac's injured list tracker. Potential regression indicator with respect to time away from playing at a high level or compromised physical performance.

Passing:

- Fantasy Pros (n.d.):
    - Target Statistics (Completions, Pass Attempts, Pass Yards, Pass Touchdowns, Interceptions): prior production key for future prediction.
    - Pressure (Hurries, Sacks): a gauge of a QB's pass blocking situation including how often they are pressured and how they respond to pressure, which can affect passing efficiency.
- NFL Next Gen Stats (n.d.):
    - Air Yards (Average Intended, Average Completed): measures QB willingness vs. ability to push the ball downfield that can drive passing yards.
    - Aggressiveness (Aggressive%): defined as percentage of all pass attempts into tight windows (defender within a yard of the receiver) potentially reflecting high risk/reward QBs who may stack up both positive (yards, touchdowns) and negative (interceptions, incompletions) counting statistics.
    - Expected Completion Percentage: completion percentage adjusted for difficulty of passes. Difference between actual completion percentage can be an efficiency regression indicator or measure of accuracy relative to the average QB if trends persist.

Rushing:

- Fantasy Pros (n.d.):
    - Target Statistics (Attempts, Rushing Yards, Rushing Touchdowns)
- NFL Next Gen Stats (n.d.):
    - Rushing Yards Over Expected (RYOE, RYOE/ATT, positive RYOE%): ability to generate more rushing yards than expected bodes well for efficiency and role stability while underperformance can spell otherwise.
    - 8+ Defenders in the Box (8+D%): reflects defenses' commitment to stopping the run by stacking the box with several defenders, which can lead to poorer rushing

efficiency. Particularly telling as to how integral the RB is to his offense and how balanced the offense is in general.

- ESPN Analytics (n.d.):
    - Team Run Block Win Rate % (TmRBWR): an offensive line's ability to sustain blocks for running plays, accounting for favorable/infavorable offensive lines that can partially contextualize rushing efficiency and volume.

Receiving:

- Fantasy Pros (n.d.):
    - Target Statistics (Receptions, Targets, Receiving Yards, Receiving Touchdowns)
    - Catchable Targets: can realistically adjust catch efficiency by discarding passes that weren't catchable.
- NFL Next Gen Stats (n.d.):
    - Separation: distance between player and nearest defender at time of pass attempt. Measure of a receiver's ability to get open and thus increase their chances of getting targeted and accumulate receiving statistics.
    - Yards After Catch (YAC/R, expected YAC/R, +/- over expected): can be telling of a player's ability to make the most of each reception by fighting and eluding defenders for extra yards.
- ESPN Analytics (n.d.):
    - Receiver Scores (Catch, Open, YAC, Overall): quantifies a player's ability to create yards from their catching, separation, and after the catch skills on a 1-100 scale. Evaluates the tools needed for players to succeed as receiving options before factoring in QB play.
    - Routes Run: number of routes run illustrates receiving targets opportunity.

To establish a recurring theme, I first conducted each aspect of this project with only RBs. I tweaked my approaches accordingly based on replicability and simplicity before settling on common methods for the rest of the positions. However, I did miss a couple of shortcuts and ways to best set up the models for accuracy. I could have pulled the Next Gen Stats (n.d.) data using nflreadpy, rather than copying from their website into an Excel file and reformatting. The data is pulled weekly in the nflreadpy (Carl & Ho, n.d.) library, but I didn't realize I could set 'week' to 0 to get the full season stats until I revisited a forgotten metric for the RBs. I also did not flag games that were shortened due to injury. The models do not see any difference between a full, poorly played game and an injury-shortened game with productive but scaled-down statistics. While nflreadpy does have individual game statistics, I would have to potentially work with data tenfold of what I am now, and would have to dictate and filter full games. Essentially I would be taking an extreme extra step of reworking season statistics into those accumulated only via full games, which was a process I felt wasn't worth it for its marginal beneficial use cases like better predicting players coming off years of multiple injuries. Furthermore, I did not include rookies as I did not develop a separate model for them, who do not have any NFL season

statistics to predict from. Currently all the models would have only been able to go off of draft picks and contracts, which is not enough to make trustable projections. That would be another project in itself involving college and NFL Combine data that I may pursue down the line, but would be separate from this one.

I had also initially planned to include Kickers in the positions of interest, although I ran into several issues constructing that dataset. Fantasy Pros, which had served me well for each of the other positions, did not include make/miss data from each field goal range (i.e. 1-19, 20-29, 30-39, etc.), instead only the makes. Since I wanted to calculate field goal percentage from each range to serve as meaningful predictors, I pivoted to ESPN. Upon copying the data into my Excel sheet, player name and team were concatenated without any spaces into between, and players traded within the season had all of their teams listed, separated by back slashes. Additionally, the make/miss data for each range had automatically formatted into month/day in Excel (i.e. 8-9 became Sept-8). I attempted to correct both in Python and had difficulty doing so, looking for other sources that had already separated makes and misses from each range into individual columns and finding no such luck. I could have written a better script to construct my desired Kicker dataset, but decided it wasn't worth pursuing as Kickers aren't a priority in draft strategy, and was satisfied with my work for the main four positions I had already built datasets and models for. That said, I will proceed to walk through the process of how I constructed the datasets.

I chose RBs as the first position to tackle due to their meaningful contribution to multiple facets (receiving, rushing), although with the purpose of only establishing standard approaches I would have been better off beginning with QBs who had noticeably less observations and metrics to include. With this, the original plan was for the data construction process to be conducted via Excel, including merging and cleaning simply via Excel functions. I dictated the main sheet in Excel to serve as the master, while creating several additional subsheets within the file to copy each new batch of metrics I planned to use. By forming a helper column chaining Player Name, Year, and Team in each subsheet, I used XLOOKUP to pull data from the subsheets into the main sheet, one column at a time. While doing this, I gradually began to realize how large my dataset was becoming, eventually identifying some 50+ variables of interest to include for a 1300+ observation RB dataset. Working with so much data through Excel on a limited machine proved to be too difficult and time-consuming, as each sequential merger of a singular column began taking several minutes. I realized this method would be far too ineffective considering I still had three other positional master sets to form, leading me to revamp my approach.

This new approach involved pulling and copying data across several individual Excel files, while using Python for the mergers and construction of the master sets. Using the RB position again as a sort of guinea pig, I initially merged all of the rushing features into one file via Excel, receiving into another, and a couple miscellaneous files. This method was further simplified into doing all

the mergers in Python for the other positions. Other than extracting player name and team abbreviation from the Player (TM) column in the Fantasy Pros' base data, Excel was mostly limited to data entry functions, rather than the all-purpose use cases I originally planned for. Thus each position's master dataset consisted of approximately 6 to 8 Excel datasets and a position-filtered misc_data.csv subset through respective {position}_master.py scripts. The misc_data.csv file includes contract, draft, and other data generated by the misc_data.py script using the nflreadpy library. I constructed the data to cover the same year ranges as previously defined (2018-present), including contracts signed before but running through 2018 before actually setting the time range for the data.

Universal data construction-related functions amongst the positions were conducted in the utils.py script. The first of two main functions of utils.py is to normalize player names that occasionally vary slightly between sources. This was done by stripping common name suffixes like Jr, Sr, roman numerals, and trailing whitespaces while removing periods and apostrophes via regex. Varying team abbreviations and some other player names (i.e. shortened first names or nicknames) were additionally renamed manually to match Fantasy Pros conventions. Players traded midseason were labelled with the teams they ended that season with, matched to a combined total of that season's statistics. With this, the data was merged through a series of outer joins, as I didn't necessarily need players to have each metric so long as they accumulated enough counting statistics, while NaNs could be treated properly later. The final merger with misc_data however required a left join, as the data pulled from nflreadpy included every player that appeared on an NFL roster, and I didn't need drafted/signed players who had had no real production. Player name, season year, and team (if possible) served as the primary merger variables. To contextualize the occasional omission of team, team abbreviation was sometimes missing in some datasets or was inputted to reflect current representation, rather than of that specific season.

2.2. Data cleaning

With the aid of Claude, data cleaning was primarily implemented in the second main function of utils.py, the volume check. This function flagged players who did not have a minimum of two adequately "productive" seasons. I created per game volume variables and defined thresholds for each alongside the counting statistic itself to account for shortened seasons and lower but consistent volume players. These thresholds served as a measure of productivity alongside a 4 games played minimum. For QBs, the minimum volume requirement was 10 pass attempts per game, or 100 total. WRs/TEs needed to average at least 2.5/2 targets per game or 28/20 total respectively. RBs were given a little more freedom and could either meet a rushing threshold (6 attempts per game or 25 total) or receiving threshold (3 targets per game or 15 total). This lenience is attributed to their volatile roles due to a harsher perceived aging curve and fiercer touch competition. Exceptions to the 2+ seasons of productivity rule were given to rookies who were productive in their only year of data, alongside players included in the starters.csv data

mentioned earlier whose projections I was obliged to include regardless of past production. Due to the leniency of the "productive" threshold, I was content with allowing players who were projected to start but with potentially lesser prior production with respect to the likelihood of similar cases in the past.

In general, I kept these thresholds pretty low considering my data dates back only to 2018 while publicly available NFL statistics span decades earlier. Additionally, I believed forecasting role stability could be realized by keeping more of the data I had. I plan on returning each year to add the previous year's data, which in due time will help alleviate this issue as I remain committed to keeping the relatively new advanced metrics in the equation. Still, a significant number of players were dropped from their respective datasets, including players whose role never materialized, one season wonders, and those whose careers ended around the beginning of my target scope. Typically around a third to a half of the players survived this check, as the original RB dataset had 414 players across 1327 seasons, of which 199 over 880 remained. QBs went from 198 players across 756 seasons to 94 over 480. WRs went from 746 over 2403 to 264 over 1234 while TEs went from 398 over 1316 to 124 over 613.

In addition, the volume check flagged leading and trailing "dead" seasons, defined as seasons directly at the beginning or end of a player's career not considered productive. This didn't drop any more players as their dead seasons had no effect on the count of their adequately productive ones, but we saw 136 seasons get dropped for the RBs, and 69, 189, and 113 for QBs, WRs, and TEs respectively. The purpose of this screening was to account for sudden, unpredictable role changes or injury-shortened careers. These would be harmful to the models due to excessively drastic year over year changes. As an example to convey the severity of this, we can take Dalvin Cook's 4 year run of 1000+ rushing yards in Minnesota coming to an abrupt stop following a yearlong stint with the Jets, in which a less favorable surrounding offense and the ascension of a young Breece Hall led to an inefficient, low-volume year that proved to be his second-to-last season. What amounted to an approximate 900 rushing yard decrease is one my models would not and should not have been able to pick up on without hindsight, posing as a confounder due to its extreme outlier nature. A case for a need to eliminate leading dead seasons would be Rico Dowdle, who recorded next to no production his first three years due to injury and being buried in the depth chart behind Tony Pollard and Ezekiel Elliot. Following a zero touch Year 3, he gained a role following Elliot's release, leading to 361 yards in Year 4 and over a thousand after Pollard's departure in year 5. Likewise to the Dalvin Cook case, this goes to show how models would benefit from the exclusion of these transitions from "something" to "nothing" and vice versa, as it lacks the context to do so.

What remains however, are dead seasons in the middle of a player's career. I could have gotten rid of all dead seasons to ease predictive ability by inputting and outputting productive seasons. For reasons as stated earlier, it would seemingly be cleaner for the models to not have to worry

about these seemingly random, staggering changes in any capacity. Yet the lack of realism of this approach led me to leaving in these sandwiched dead seasons for explicit reasons. The prevalence of role volatility is meant to be respected particularly in a high-turnover sport like football. Decline is inevitable yet relatively unpredictable, and leaving only productive seasons would make this even harder to project. While it is not easy to predict these gap seasons, they could at the very least serve as the beginning of the end, a blip on the radar that may turn into something more meaningful. What particularly enables me to allow these gap seasons is the fact that a player's role as an adequate producer has already been established. The trajectory has already begun. Leading and trailing dead seasons precede and succeed this "trajectory" and are thus discardable. It is unrealistic to be able to predict a prolonged meaningful stretch of "something" after beginning with a period of "nothing". Once a player's role as an adequate producer is done, there is no need to use their remaining data especially when we are predicting players who are expected to still be productive. Put simply, leading and trailing dead seasons are not "tells", but those in the middle of a player's career can be. And typically, these gap seasons were accompanied by far less statistical change due to a player's establishment as a starter or meaningful producer. Since the models are built primarily on per-game statistics, even injury-shortened seasons were comprised of reasonably productive games given that the player already had a defined role.

Both minimally productive players and leading/trailing dead seasons for each position were outputted to separate csv files for an inspection in which I tweaked the volume thresholds. Typically I had started with higher thresholds and checked the dropped players to see if they were indeed reasonably droppable. I gradually bumped them down as far as I could without dropping anyone too significant, ensuring I maximized the usable data for the models to work with. As mentioned earlier, as I continue to add more data over the years, I will slowly be able to increase the thresholds accordingly again to provide greater security against barely eligible, low-volume players sneaking in.

2.3. Preparation of Data for the Models

This step mostly entailed the lagging of certain statistics, converting counting statistics to per game statistics, and some other feature engineered columns of my choice within the build_features function in base_prep.py. The former two were defined in position-specific lists housed in each position's feature map, which will be touched on later. As for the feature engineered variables, Years Since Productive was written in the utils.py script for simplicity as the vol_check function was already flagging productive and dead seasons. The base_prep.py script contained the rest of these features, including a similar Years Since Full using 13 games played as a "full" threshold. These particular features were meant to serve as regression indicators, accounting for rust associated with prolonged time since a meaningful role or from playing in general. Percentage of Peak Role was implemented by calculating a player's season high number of touches (passing attempts for QBs) up until that point of a player's career, while

dividing the given season's number of touches to this prior max to prevent leakage. Thus players were able to achieve a greater than 100% value if they set a new career high in touches. While the previous features were more indicative of recession, this one was able to show signs of role growth alongside decline.

Undrafted players received a value of 300 for draft pick and 8 for round for a diminished quantification relative to drafted players, but serving better use than a NaN. Other simple features include Years of Experience, Team Change, Games Missed Rate, and Age/Age Squared. 3 year-prior standard deviations were calculated for volume based statistics like attempts and targets to aid in the accounting for role instability. Some position specific engineered features include INT% (interceptions divided by attempts) for QBs and xCatch% (receptions divided by catchable targets) for WRs. As for the lagging conventions, lower volume, volatile statistics were generally lagged by two seasons, while most of the rest were lagged by a single season. All of the target statistics were lagged by three seasons for smoothing to lead to better predictions. After all of this, I was left with a set of 153 features for 172 QBs, 133 for 209 RBs, and 150 columns for 423 WRs and 291 TEs.

Now armed with all the features I wanted, I constructed a target feature map for each position while combining WRs and TEs into one, as they had the same features across the board. This was done to cater towards Lasso, GBM, and MixedLm, which do not zero coefficients of parameters. I didn't want them to fit to the noise that may be associated by flooding them with access to all the features, as many of them have little to no correlation with certain target statistics (i.e completion percentage and rushing yards for QBs). Within a respective `{position}_map.py script`, feature maps were defined as dictionaries with the keys being the target statistics, and the values being lists of relevance features used to predict said target statistics. Alongside the aforementioned counting and lagged statistics, each `{position}_map.py script` defined lists of features that could be used to predict multiple target statistics within the position. For example, `RB_map.py` script defined cross, base rushing, and base receiving features. Cross features were applied to each target statistic's mapping of each position and included draft, contract, age, and some snap share data if relevant. Rush Base was applied to each rushing statistic and Receiving Base to each receiving statistic, while the touchdowns for either were padded with their respective Redzone statistics and adjusted touchdown rates. This general approach held true for each of the other positions, with the base statistic types (passing, rushing, receiving) varying respective to the position.

Handpicking the features for each helper list and target statistic in general went through a couple iterations. I didn't want the models to pick up on seemingly apparent trends in certain statistics and reflect them on my projections, so I aimed to keep each target statistic's mapping as relevant as possible. I started with compiling every remotely similar metric to its respective target statistic and slowly began to drop less valuable features while keeping an eye on the root mean squared

errors for each model (I had already written the scripts generating the projections by this time). For the most part, this went smoothly outside of a particular issue that arose after peeking at the 2026 projections, in which almost all QBs were originally predicted for between 10 and 12 interceptions. I suspected they had been somewhat normalized across players due to the cross features being somewhat less correlative (i.e. age, draft pick), leading me to making a specific mapping list for interceptions. Now based more on interception, accuracy, and pressure-related statistics, rather than passing in general, I was able to bring reasonable variability back and prevent everyone from regressing to the mean.

3. Analysis

3.1. Model choices + implementation

With experience in Lasso, Ridge, and Random Forests in the programming language R, I opted to train the former two alongside two unfamiliar methods that made sense for this data: Gradient Boosting and Mixed-Effects. These were two of several suggested by Claude. I chose Gradient Boosting because of its familiar decision tree-based learning, yet being more advanced than Random Forests with its ability to sequentially train decision trees to predict the gradients of the loss function, rather than the labels themselves. Mixed-effects caught my eye due to its ability to identify players themselves through intercepts, which allows me to predict using the context of a player's entire career, rather than just predicting a season from a couple of prior seasons. This was more along the lines of what I expected to be a prototypical machine learning model for player careers, allowing the model to grasp the capabilities of the players themselves, and be able to notice how in line certain seasons were relative to career average or expectation. For my project, this seemed particularly attractive for players with more than 3 years of data, as that was the furthest my statistics had lagged.

Each model had similar methods of implementation, which was done in the `fit_and_pred.py` script. Helper functions for fitting and predicting each were written here. Scalers when present were standardized with StandardScaler. In addition to sklearn and statsmodels' documentation pages, I used Claude's guidance for a significant portion of this process, particularly for LightGBM and MixedLM whose concepts I had less of a grasp of. With this, a conventional 5 fold cross validation was used for Ridge, which tried 30 penalty parameter alphas spaced logarithmically between 0.01 and 1000. Lasso similarly used 5 fold cross validation, but the alphas ranged from 0.001 to 10 with respect to the potential of its L1 penalty zeroing every coefficient after exceeding its alpha max. Furthermore, the number of maximum iterations was increased to 2000 from sklearn's default of 1000 to better ensure the model could converge cleanly due to Lasso's coordinate descent. For LightGBM, I additionally scanned through Sarah Zouinina's guide on hyperparameter tuning. The number of boosting rounds was limited to 500, although the conservative learning rate of 0.03 (due to lack of training data) and early stopping

after 30 consecutive rounds of no improvement contributed to the number of estimators falling much short of this max to prevent overfitting. The max splits of the trees was a modest 4 due to a lot of collinearity between several of the included features reducing the data's overall complexity. The number of leaves was accordingly 15 (2^4 - 1) and the minimum number of child samples was 10. Subsample (enabled by subsample_freq = 1 and re-sampled each round) and colsample by tree were both set to 0.8, so each tree was trained on a random 80% of the rows and another 80% of the features. Finally, MixedLM essentially ran with the defaulted or standard parameters, utilizing the prototypical Restricted Maximum Likelihood to estimate variance components with less bias. The defaulted L-BFGS optimizer fine-tuned the parameters, while the max number of optimizer iterations before giving up was originally set to 200.

Most of the remaining fit_and_pred.py script entailed an evaluation and prediction function predicated on finding the model best at minimizing root mean squared error, in which Claude provided a bulk of the programming. First established was the subsetting of existing data into rows with known next-season targets. The list of players to predict were built from prior season data and filtered to those whose last season is the current season (indicative of activeness). I had constructed the starters.csv dataset after completing this script, in which changes were implemented in the model_run.py file, subsetting each master set to players whose gsis_id were included in starters.csv there as opposed to altering fit_and_pred.py. From here, a helper function extracting coefficients and feature importances from the models was defined. This included both scaled and raw coefficients, although the latter was not utilized.

This gave way to the per-stat loop that fitted the models and generated the projections. To prevent information leakage, the training set was defined as the years 2018 through 2023 while 2024 served as the validation set, rather than a random split. For the models that couldn't handle NaNs (Lasso, Ridge, MixedLM), the imputing step filled missing feature values with the median of the non-missing values in the training set, as filling with zeroes wouldn't make any sense here. The missing feature values were relatively extensive due to advanced statistics requiring a much more significant volume threshold than I allowed in my data. Furthermore, lagged statistics for non-existent seasons (target statistics were lagged 3 seasons) and percentage of peak volume prior to the peak contributed to the collection of NaNs. With this addressed, each model was fitted on the training set and predicted on the holdout as defined earlier, with both being stored in a resulting dictionary. Mean absolute and root mean squared errors were calculated for each model, in which the two models best minimizing the latter were marked. The second was a backup in case the first model didn't converge (which happened to a couple TE iterations in some of the run-throughs). Otherwise the first became the final model for each specific target statistic and position combination, whose coefficients/importance measures were then extracted. This concluded the per-stat loop, in which the rest of the function began constructing four objects to return. This included the actual projections by position (which added a transformation into fantasy points via another helper function), the best models for each statistic/position

combination, comparisons of each model's errors, and the coefficients/importances of each feature.

3.2. Model Evaluation

Each position besides QBs had 7 target statistics, while QBs had 8. The following displays the distribution of model wins across varying subjects. To further note, these visuals were implemented in visuals.ipynb using the matplot library, lightly based on some work in previous classes and refined with the aid of Deepseek.

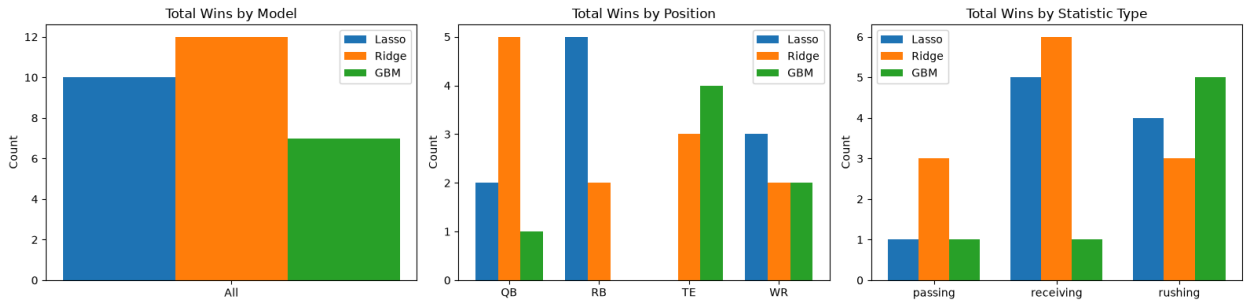


Table 1: Selected Best Model Comparisons

The notable MixedLM omissions were due to the model converging on only 8 of the 29 total position-specific target statistics while the other models converged on each one. Furthermore, it finished dead last, typically by a significant margin, in each. This remained constant even after I changed the max number of iterations from 200 to 300 and eventually 500. This can potentially be attributed to the limited amount of data I worked with, considering I could only set aside a year for the validation set. Another factor is its redundancy with the lagging and inclusion of the target statistics in explaining its inability to identify random intercepts. That aside, 7 of the 8 were notably rushing statistics, with rushing attempts, yards, and touchdowns converging for both QBs and RBs. This could explain rushing ability being a stable player trait, better identifying dual threat quarterbacks and workhorse running backs from pocket passers and complementary backs, respectively.

Referring back to the table, of the remaining models, Ridge and Lasso each comfortably outpace or match GBM across all disciplines aside from TEs and rushing statistics. Ridge especially seems to be the best all-around model, performing the best in passing and receiving disciplines while maintaining pace in rushing. It also remains significantly present in each position, while Lasso (TE) and GBM (RB) completely drop out of one. GBM finished first in 5 rushing statistics between TEs and WRs, interceptions, and TE receiving yards. A common theme among these is that they are lower volume and had less relevant features to work with, so that seems to be GBM's strength for the particular data it worked with in this project. It was able to make the most of The rest of the position-specific target statistics went down to a close split between Ridge and Lasso. Ridge particularly had a stronghold on QB statistics, winning each besides

Completions and Interceptions while also winning 3 of the remaining 4 TE statistics. Lasso served as the primary model for RBs, winning each non-touchdown statistic, while also taking WR receptions, receiving yards, and rushing touchdowns. From this, we can also generally observe that Ridge performed best at touchdowns at positions with meaningful counts, as the only touchdown statistics it didn't win were rushing for QB/WR/TE. These can be pretty low-volume relative to the ones it did win (QB passing, RB rushing, RB/WR/TE receiving). This approach of fitting four models for each characteristic and picking the best one was also super interesting in seeing the strengths and weaknesses for each given the same data.

Overall, the linear models took the cake, winning 22 of the 29 total statistics. I will say, due to my knowledge of football, I was able to craft relevant feature maps that particularly aided Lasso (which was the reason why they were created in the first place), giving it an advantage. Given that Ridge won the most target statistics anyways, it seems that they followed more linear trends anyways. But going back, I definitely could have applied the feature maps just to Lasso, given Ridge and GBM access to all the features, and rewritten MixedLM-specific maps excluding lagged and target statistics. I will probably test this out later, but I've written too much of this paper catered to the model-wide feature map application to delve into that within this. In general, I have quite a bit of work to fix MixedLM, and will probably get around to it periodically as I focus more on school now. I definitely didn't originally have the best understanding of how the model worked, and have continually pinpointed the issues that can be worked out through better approaches. These changes will likely be reflected in the GitHub repository a bit before I revise a second version of this paper with hopefully meaningful MixedLM projections and findings to share.

3.3. Statistical Coefficients and Importance

For this step, I combined the target statistic type across all multiple positions, ranking the 5 greatest parameter coefficients in both directions. Since Ridge models zero irrelevant features and BGM evaluates them by importance, they instead essentially just included the top 5 instead. Positions were grouped into QBs/RBs and WRs/TEs. This ended up creating some graphs within the loop that don't make sense (i.e. QB/RB int), but the main purpose was to compare parameters from comparable volume statistics shared by certain positions, like rushing statistics for QBs/RBs and receiving for WRs/TEs. This also condensed the plot down from the original 29 total target statistics to 21 individual graphs for moderately improved readability. This partially alleviated the case of a statistic having a GBM and a linear model winner, which originally complicated 3-4 of the original graphs due to the scaled differences between the two parameter interpretations. A better fix was later implemented that automatically separated position groups whose statistics were won by models with different parameter evaluation types.

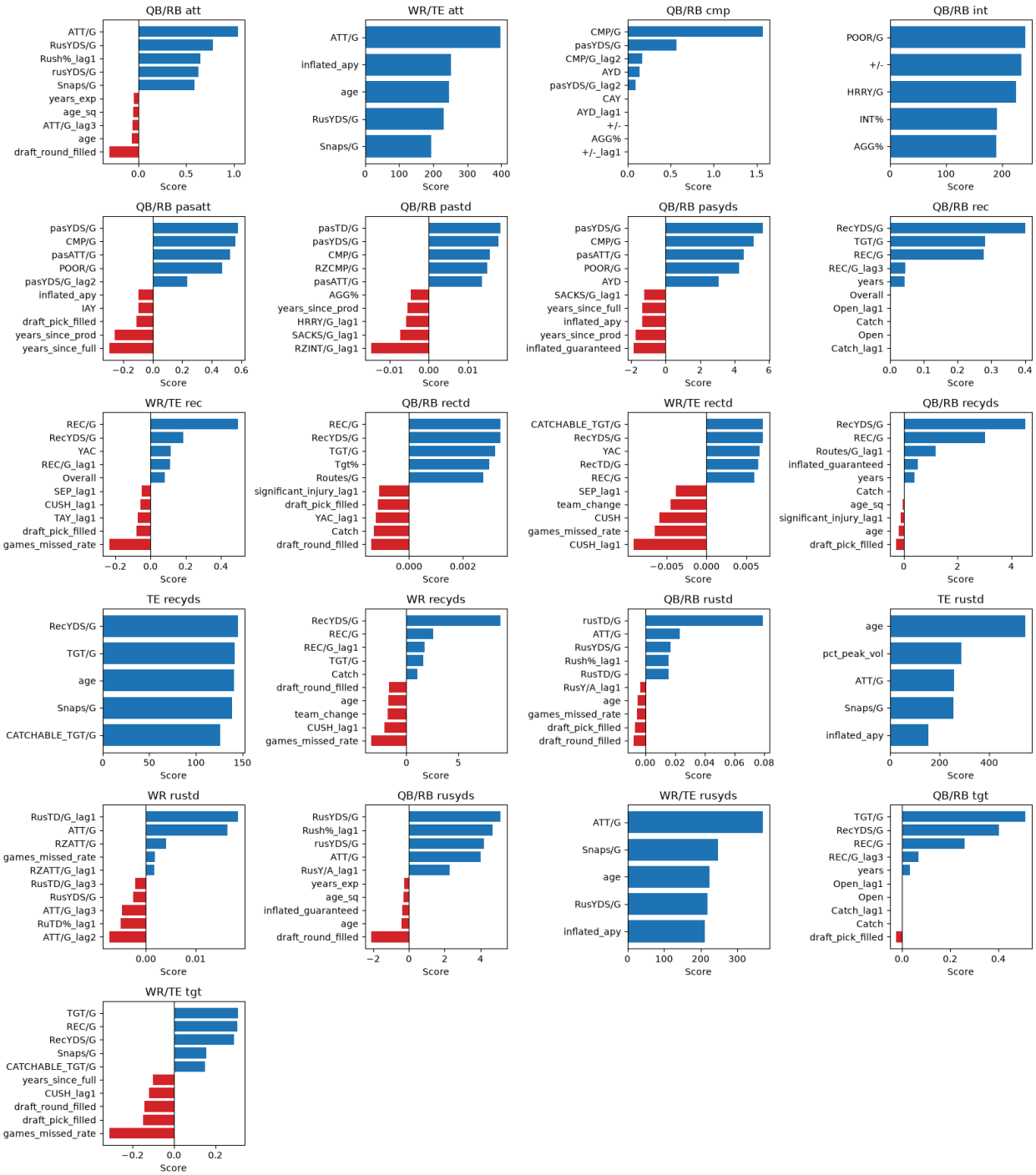


Table 2: Strongest Coefficients Comparisons

This was one of the most interesting steps of the project, which allowed me to interpret correlation between the parameters and target statistics. Drawing these connections allowed me to cross check my findings with my prior sort of football common sense and logic commonly shared among people who follow the game. For the most part, the ranked relative magnitudes of the coefficients follow typical expectations of associations. Typically one of the greatest positive

parameter coefficients was the target statistic itself along with other target statistics within that subject, which all directly influence each other. Yards and touchdowns were driven strongly by opportunity, as the ability and developed trust to get the ball typically gives players chances to make plays at higher rates at all stages of the field, including the endzone. Age and age^2 placed highly in almost every non-passing statistic, reinforcing the harsher perceived aging curve of skill-position players, as opposed to quarterbacks who typically have longer career spans due to their production being more predicated on arm and cognitive ability, which deteriorates at a lower rate than general athletic ability due to the lesser physical impacts of the game in those aspects. Another relatively common theme was the inverse relation between draft status and many of the opportunity-reflecting statistics (passing/rushing attempts, receptions, targets). As this holds true for a dataset including a wide range of older and younger players, it shows that NFL teams typically do a pretty good job at drafting and developing players. This makes sense in explaining how coaches are more inclined to trust higher drafted players with more opportunities due to their potential stemming from projection out of college.

In general basic metrics seemed to hold the greatest weight in the models, with the advanced ones appearing more sparsely throughout the graphs. This is likely just a product of how common basic metrics were in the data, and how the advanced metrics themselves couldn't separate themselves at a higher rate than the basic ones. With this, interceptions and WR/TE receptions and receiving touchdowns were the only graphs in which advanced statistics made up at least a third of their displayed features. Interceptions in particular were actually mostly driven by advanced statistics, as the winning GBM model found the parameters poor passes per game, completion percentage above expected, hurries, interception percentage, and aggression percentage among the most important. It was interesting how passing attempts (which finished 6th) were surpassed by all of these despite being top 4 in every other passing statistic. This shows how interceptions are more of a product of pressure, inaccuracy, and aggressiveness relative to accompanying high volume.

As for the receiving statistics mentioned above, yards after catch, targeted air yards, catchable targets, cushion, and one year lagged separation made the cut in the graphs, with the first three appearing as positively correlated with their respective target statistic and the latter two being negative. Separation being negatively correlated with receptions and receiving touchdowns stands out as an odd finding in this data, given that a player's ability to separate himself from defenders at the very least should be a neutral look for their ability to catch the ball and score. A plausible explanation could be that separation bears more influence on target opportunities, while actually receiving the pass is a different story. This potentially could illustrate an inverse relation between the two as the negative coefficient suggests, but might just be the model fitting noise or having a lack of data (considering the separation statistic has only been around since 2018). Otherwise, the remaining advanced statistics made sense for mostly obvious reasons, as receiving yard and touchdown accumulation is driven by making the most of catchable targets,

aided in them being further downfield and ability to create more after the catch. Cushion is something I didn't discuss in the Data section because I thought it wasn't worth mentioning, but ended up playing a decent role in some of the models. It is defined as the distance between a receiver and the defender they are lined up against at the time of the snap for targeted plays. We see a negative correlation between this and receiving statistics as lesser cushions primarily played through press coverage can throw off receiver timing and route running, as defensive backs are allowed to jam receivers within the first five yards. We see this magnified for touchdowns (cushion and its one year lagged statistic have two of the three strongest negative coefficients) as there is much less space to work with in the Redzone to create space. While on the topic of touchdowns, touchdown percentage and expected touchdown percentage did not make an appearance in any of these graphs, in which the absence of the latter was surprising given my initial high expectations for it. It seems that touchdowns are too volatile for these statistics to make many waves, relative to opportunity and the touchdown counts themselves.

4. Results

The full list of projections can be found in the projections folder. In this section, I will be discussing the 2026 projections, which is relevant to the time I am writing this report. I made copies of each position's projections and used Excel formulas to extract means, medians, etc. of varying ranges of players to discuss my findings.

The top 5 projected QBs in order were Josh Allen, Patrick Mahomes, Jalen Hurts, Jayden Daniels, and Trevor Lawrence. Contrarily, the bottom 5 beginning with the least projected productive quarterback were Aaron Rodgers, Kirk Cousins, Malik Willis, Geno Smith, and Tua Tagovailoa. Both of these lists seem completely reasonable given top 5 QBs are highly productive on the ground (Lawrence surprisingly rushed for 9 touchdowns last year while the rushing ability of the others has been established) and reasonably so in the air, reflecting the dual-threat status that has become the standard of a modern fantasy football quarterback. Meanwhile the bottom 5 are expected to command lowly offenses for poor teams. The model seems like it's doing a solid job in the rankings, with only a few players in relatively odd spots I will mention later. Upon first glance however, there are some questions about the ceilings of these QBs, as Josh Allen's position-leading 2026 projection of 20 fantasy points per game was surpassed by 6 QBs in 2025. One of whom, Matthew Stafford, is expected to experience a huge regression from 21.1 FPT/G (2025) to a projected 15.9 FPT/G for 2026. He led the league in passing yards and touchdowns with 4707 and 46, respectively, yet is projected to decline to 4250 and 30, which is somehow still expected to lead the crop of quarterbacks for 2026. These would be the lowest for a leader since 2005 for passing yards and 2002 for passing touchdowns.

A notable outlier player is Malik Willis, who is entering his first year as a Week 1 starter with very limited production through full and partially played games. His projections are all screwed up as a result, but the 30 remaining quarterbacks (Cleveland had not officially updated the depth chart at the time the data was pulled) were notably projected for between 3520 and 4250 passing yards and 23 to 30 passing touchdowns. Of QBs who played all 17 games in 2025, the corresponding ranges are 3169 (Cam Ward) to 4707 (Stafford) and 15 (Ward) to 46 (Stafford). However, removing the minimum and maximums of both gives us 3668 to 4564 and 23 to 34. Comparing the medians, we have 2025's 4007 passing yards and 27 touchdowns to the 2026 projections of 3709 and 27. As a whole, the crop of 2026 QBs are projected to average 16.5 fantasy points per game, while the top 32 QBs of the 2025 season averaged 17.45. While the Stafford case seems to suggest a staggering regression to the mean, a closer look at the rest of the distribution seems to suggest it has changed rather minimally, although in the negative direction. With hindsight we can definitely say that several players last year outperformed expectations, maybe more so than those who underperformed which can potentially explain the collective dropoff my model seems to expect. Meanwhile the rushing statistics also held over relatively well from 2025. Since quarterback rushing ability comes with more polarity than passing, I will only compare the medians, which come to 2025's 356 (only including QBs who played 17 games) to 2026's projected 288.

However, I noticed a trend amongst prototypical pocket passers who offer less rushing ability. Relative to other productive quarterbacks, they were disproportionately flagged for general regression with Jared Goff, Matt Stafford, Joe Burrow, Dak Prescott all falling outside of the top 15 in fantasy points per game despite the opposite being true for last year. Their average age is 32.3 years, which is much higher than the 2026 crop's average age of 28.4. Joe Burrow actually dragged this average down, with him being 28 and the rest in varying stages of their 30s, although he particularly had a down year (by his standards) in 2025, which was why he may have been part of this group. In general though, there haven't been many signs in reality of them slowing down given their balanced offenses and willingness to stretch the field. The model however wouldn't be able to see that, and is potentially reading into their increased age as a significant regression indicator. Otherwise passing statistics may have taken more of a hit regression-wise than rushing, even though they tend to be more stable with age. As a whole though, I found the QB projections very reasonable and in line with industry standards.

The RB projections including 56 players are topped by Christian McCaffrey, Bijan Robinson, Jahmir Gibbs, Jonathan Taylor, and Saquon Barkley. From this point onwards, listing the least 5 projected players is less meaningful as they share touches with the rest of the skill position players, as opposed to a team's starting quarterback expected to take every snap. It would be unfair to compare given their drastically lesser workloads. In 2025, 17 RBs eclipsed 1000 rushing yards, and that number holds true for these 2026 projections. The highwater mark decreased from 1621 (James Cook) to 1476 (Derrick Henry), while the rushing touchdown high

of 18 (Jonathon Taylor) fell to 12 (Derrick Henry). While the touchdown decrease seems significant, we only need to date back to 2015 to find a high that low, and rushing touchdowns are pretty fickle and particularly hard to predict a maximum. The top 32 RBs were projected for between 838 and 1476 rushing yards and 6 to 12 rushing touchdowns. Receiving yards stayed relatively steady as well, with the two leaders (Christian McCaffrey and Bijan Robinson) regressing a bit from their 800+ receiving yard 2025 seasons to the mid to high 600s, but otherwise the rest saw marginal declines for the most part. Receiving touchdowns however had a narrow range, with one of the 56 players projected for none and the rest between 1 and 3, while 8 RBs surpassed that number in 2025. This is likely due to RB receiving touchdowns being a low-volume statistic with quite a bit of noise, so this makes a little more sense than passing touchdowns and yards similarly being tightly restricted to a range despite their larger values to work with.

My models were not handed information indicative of significant role changes related to changing touch competition, as I was unable to implement a sort of expected metric for that, which was originally one of the things I was most inclined to do. As I have mentioned earlier, this project is not a finished product, and I can and will improve upon it in the future, although what remains in my capabilities will define what this project is now. Additionally, 2026 projections also matched players with their 2025 teams, so I noticed a few shared backfields and players who changed teams projected for lead roles with unexpected projections. Some of these team changes left the player's role unchanged, such as Travis Etienne Jr. being expected to garner a similar amount of touches in New Orleans as he did in Jacksonville. However, the model's projected Jaguars backfield reflected Etienne as its leadback, as opposed to Bhayshul Tuten who was expected to mostly fill that role alongside Chris Rodriguez Jr. in a secondary role. We thus have Tuten projected for just 634 rushing yards despite being the expected RB1 of the Jaguars. Some other backfields affected by similar situations include the Saints, Panthers, Texans, Seahawks, Chiefs and others, reflective of this year's free agency resulting in a lot of new faces to teams. I will particularly highlight the Panthers and the Seahawks/Chiefs. In 2025, the Panthers' backfield was led by Rico Dowdle, with Chuba Hubbard, the reigning RB1, regressing significantly and falling to the RB2 role. In 2026 Jonathan Brooks will return from missing essentially the entirety of the last two seasons, yet is projected to lead the now Rico Dowdle-less backfield with 934 rushing yards despite having next to no data to predict from. While many fantasy analysts are forecasting similar projections, this is pretty baffling because my model doesn't really have any context other than him being a younger second round pick. To touch on the Seahawks and Chiefs backfield changes, we have Kenneth Walker with very muted projections (929 rushing yards) despite now having the lion's share of Kansas City rushing attempts to work with. Since their 2025 primary RBs, Isaiah Pacheco and Kareem Hunt, are no longer with the team, the 2026 RB projections actually bear no representation to the Chiefs. What these examples have magnified is the lack of context the model has in its ability to reflect role changes due to non-numerical statistics. I have quite some work to do in implementing team

changes to the target year projections and forecasting role itself, as opposed to predicting it entirely based on past production. However, I think the model did a pretty good job in using history to predict the future while maintaining the hierarchy of running backs well.

Now discussing the 106 player set of WRs, we have a projected top 5 of Puka Nacua, Jamarr Chase, Jaxon Smith-Njigba, Amon-Ra St. Brown, and Ceedee Lamb, the top 4 of which holdover from 2025. Themes of this dataset are pretty similar to the RBs. 18 WRs finished with at least 1000 receiving yards in 2025, 22 projected for 2026, which is not an unfathomable jump. There are some reasonable declines in the position leaders for receiving yards and touchdowns, from Ngigba's 1793 and Davante Adams' 14 to Nacua's projected 1524 and Chase's 10. Some team changes that affected projections are again present here, but aren't as impactful as wide receiver groups get more opportunities to go around. This can be seen with Malik Washington expected to lead Dolphin's pass catchers with 593 receiving yards if we don't count now Denver Bronco Jaylen Waddle's projected 1089. Emeka Egbuka is projected for a somewhat low 934 with Mike Evans gone, who himself is projected for a massive decline to 657 receiving yards after famously reaching 1000 his first 11 seasons before various injuries prevented him from doing so in 2025. Once again, we see the model particularly listens to age and recent year over year declines as signs of significant regression.

What is to note is that the model is more believing in younger first round picks with brief stretches of success, including Quentin Johnston, Marvin Harrison Jr., and Brian Thomas Jr. They are projected for between 863 and 951 receiving yards despite their 2025 receiving yard counts varying between 608 and 735. The ESPN general receiver scores (54 for Harrison and Johnston and a measly 18 for Thomas on a 1-100 scale) also suggest the talent hasn't translated as well, although that statistic didn't gain much traction in the model anyways. This does reflect the importance of age and draft status to potential, but these players have been declining or being unable to build upon prior seasons despite good health. A slighter note is the model essentially force-feeding players with receiving touchdown counts. The lowest projected wide receiver, Tay Martin, is projected for just 5 receiving yards and somehow 3 receiving touchdowns, with 2 receiving touchdowns being the lowest mark among all WRs. While he is an extreme, as 104 of the remaining 105 players were projected for at least 294 receiving yards to somewhat justify the 2 touchdown count minimum, the presence of his case in the first place seems to suggest that the winning receiving touchdown's model (Ridge) seems to default to regressing to the mean for extremely low-volume players. Otherwise, there really isn't much to say, as the projections seem very realistic for most of the remaining players, and unlike QBs and RBs we can only meaningfully draw connections for one aspect of the game (receiving) rather than two.

Finally, the 55 player TE set has a projected top 5 of Trey McBride, Brock Bowers, Sam LaPorta, Travis Kelce, and Dalton Kincaid. A somewhat reasonable top 5 but this particular position seems to have some serious inconsistencies with industry expectations. The model

doesn't seem to be a big believer in TEs with less than 3 years of experience, with Bowers (855 projecting receiving yards), Tyler Warren (711), Harold Fannin Jr. (609), and Colston Loveland (416) seeing moderate to massively underwhelming projections for 2026. Each is projected for a decline despite their youth, lack of injury history, and productive first seasons, while 3 of them were additionally first round picks. Bowers actually gets drafted in fantasy football leagues ahead of McBride a lot of the time (who is reasonably projected for 1257 receiving yards), but is somehow projected to fall a massive 400+ receiving yards and 4.6 fantasy points per game behind. Warren, Loveland, and Fannin are second year players who immediately established significant roles their rookie season, yet the model expects them to lose out some touches while the slightly older LaPorta, Kincaid, and Kyle Pitts are projected to maintain or even increase their workloads. Loveland's case is particularly baffling. While the model can't see his late season/playoff surge, he is projected for a 300 yard decrease in his second year as an early first round pick, falling from 14th in fantasy points per game in 2025 to 27th in the projections. AJ Barner has only one more year of experience (same as Bowers with 2), was a fourth round pick, managed 519 receiving yards in 2025, yet is projected to outperform Loveland.

Something to note is that TE receiving yards was won by GBM, which hasn't represented any other target statistics I've meaningfully talked about in this section besides the brief mention of interceptions. My first suspicion was model overfitting, in which the models were trained on 2018-2023 data and validated on 2024, before 3 of these case mentions began their career, while Bowers didn't have a second season under his belt yet. If we take a deeper look, within the range of 2018 and 2023 (notably excluding 2024 as there would need to be a next target season), there were 8 first or second year TEs with at least 700 receiving yards. LaPorta, McBride, and Jake Ferguson in 2023, Pat Freiermuth in 2022, Kyle Pitts in 2021, TJ Hockenson in 2020, Mark Andrews in 2019, and George Kittle in 2018. The group collectively averaged 898 receiving yards their first productive (700+ yard) season, and 671 the following, a staggering 227 yard decline. McBride was the only one of the 8 to even see a jump (825 to 1146). Remove him from the subset and we see a 305 yard average year to year decline, or 603 down from 908. This also isn't solely a product of poor injury luck either, the model predicts from per game statistics while each played at least 10 games. This is real information the model is getting fed, and with only one exception (McBride) to the TE sophomore/junior slump, the model essentially has no choice but to flag these types of players for decline. This is a super interesting find, as I was very confused why the model was so pessimistic about players with seemingly bright futures, but upon looking at past data myself, I was able to pinpoint the model's reasoning. This could be a trend worth acknowledging moving forward, but I believe it happens to be a product of the limited data I have and is representative of the "good data in, good data out" rule. I would love to see Loveland, Fannin, and Warren break these trends, and it will be interesting when I check in next year to add 2026 data to see if this prediction holds true or not, and how my projections help up in general.

5. Discussion and Conclusion

This project opened my eyes towards the intricacies of the data science world. I had learned how modeling took only a small fraction of the process and experienced it firsthand after spending so much time on the data cleaning and feature engineering steps, trying different approaches to ensure the data was adequately ready to be fitted on. Through trial and error, debugging, confusing results, and everything in between, I've learned to be resilient, to be persistent in identifying root causes while searching for solutions. I have more work to do to improve the model, and am still motivated to do so even after setting this project down for a bit to focus more on school. I've learned to trust my intuition, because my perseverance will allow me to come out on the better side. I've also learned to grasp my own capabilities and understand the gap between who I am today and who I can be tomorrow. Utilizing AI has accelerated my learning process and exposed me to much more than I could have asked for. Having a mentor with all-world knowledge on call 24/7, at every hour I need it, has truly allowed me to make the most of this project. Writing this now 20+ page paper has really affirmed how much I was able to absorb and integrate into my knowledge. In full transparency, I must thank AI for the strides I was able to make, from having basic machine learning knowledge to being able to design a full end-to-end pipeline. I always knew what I wanted to achieve with this project, but couldn't see it through solely by myself.

If you're still reading, then you could maybe say that I did abuse AI in the conducting of this project. Maybe from what I've described, I relied on it too much, maybe I gave myself a shorter leash than I needed to and fell back to AI when I could've spent more time learning. Maybe I would have benefited more if I did this all myself and learned the processes from the ground up. But I realized there wasn't enough time to get from where I was to where I wanted to be. I am entering a crunch time that is significant for the prospects of my career, and I knew I couldn't make the jump I wanted by myself. I needed to fasttrack this journey, and turned to AI to do so. I truly believe I utilized AI as a tool, and not a crutch, but that will remain to be seen as I take my next steps after completing my undergraduate degree, with this project as a launchpad. Sure I can define this project as a representation of my skills or attention to detail. But my biggest takeaways have been acknowledging my humility and growth.

Following my Junior year, I got rejected from all of the internships I applied to despite applying to several more than the previous year, which had the same result. What I didn't realize was that I hadn't prepared myself enough to be considered a qualified job applicant, as someone who could be trusted in a field that requires expertise. I hadn't set out to separate myself outside of the classroom, I really wasn't putting myself out there in the first place. It was and still is a humbling experience, and it made me realize that the growth I needed was part of a journey I could begin myself. I had always thought the first step was enrolling in university, succeeding in my courses, and hoping someone would notice. I have maintained a GPA of 3.9 as Combined

Economics/Mathematics Major and Computer Science Minor, worked as a Peer Advisor and eventually promoted to Lead Economics Peer Advisor for UCSC's Economics department, and yet I have no internship experience under my belt. But this whole time, I was only doing what was in front of me. I was excelling, but what does it matter when the world is full of the unexpected, and the unprompted?

This realization led to the idea for this project, combining the learning process of industry-ready skills with a personal passion I knew would keep me engaged. I remember that first day, I sat down and began researching basic and advanced football metrics, copying over the data into Excel spreadsheets and trusting it would all come together. There was so much I didn't know I needed to know, so many moving parts whose existence I didn't know of. The vision from that day has changed so much since then, but has remained my own. I have constantly been brought back down to Earth, learned to be realistic, but also learned how to make "realistic" something that could grow as well. Looking back has made me realize that I have always been constrained to what seems "realistic" to me, but not how I could change that. The growth that I set out to pursue, and the growth that I have made has made this change possible. Is this model usable for fantasy football players? There are many better, free alternatives including the internal projections within whatever platform they may use. Will I even use this model in my own draft strategies? I still trust my gut and knowledge of football above all, while ESPN and Sleeper rankings will continue to serve as secondary tools. I didn't set out to beat out the industry leaders, nor did I set out to become the next machine learning engineer. The plan was to be me, through and through, but a better version, one that is more ready for the next step. It wasn't the plan to become introspective in the final pages of this paper, but looking back has made me appreciate how important this project has been to me. It has given me an initiative to take control of my career. It's not about who I am today, it's who I have come to be as a product of my past. I am projecting growth for myself in these coming years, and that's not just based on past trends. This prediction is predicated on conviction and commitment.

References

Anthropic. (2026, October 17). *ml ff* [Large language model chat] Claude. https://claude.ai/project/019f1f17-0a0e-749e-a3ef-17aebdd82c6c

Carl, S., & Ho, T. (2025). *nflreadpy: Python port of nflreadr* (Version 0.1.5) [Computer software] https://github.com/nflverse/nflreadpy

DeepSeek. (2026, October 15). *FF ML Project Visuals* [Large language model chat]. https://chat.deepseek.com/a/chat/s/cc14ba41-8ad3-4fec-92a2-076d4fcd306f

ESPN Analytics. (n.d.). *NFL Receiver Scores*. Retrieved August 4, 2026, from https://espnanalytics.com/receivers

FantasyPros. (n.d.). *NFL Research Discover*. Retrieved August 4, 2026 from https://www.fantasypros.com/nfl/discover/research/

National Football League. (n.d.). *Next Gen Stats*. Retrieved August 4, 2026 from https://nextgenstats.nfl.com/

Spotrac. (n.d.). *NFL Reserve List Player Tracker*. Retrieved August 25, 2026 from https://www.spotrac.com/nfl/injured/_/year/2026/scope/cumulative/view/player

Zouinina, S. (2024, October 28). *A deep dive into LightGBM: How to Choose and Tune Parameters*. Medium. https://medium.com/@sarahzouinina/a-deep-dive-into-lightgbm-how-to-choose-and-tune-parameters-7c584945842e